Lecture 2: Some Useful Signals

CSE 219: Signals and Linear Systems

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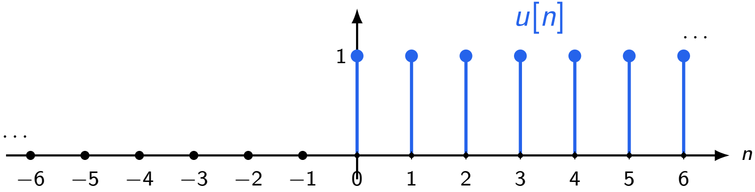
Unit Step and Unit Impulse Functions

The discrete-time unit step turns on at $n = 0$

$$u[n] = \begin{cases} 0, & n < 0, \\ 1, & n \geq 0. \end{cases}$$

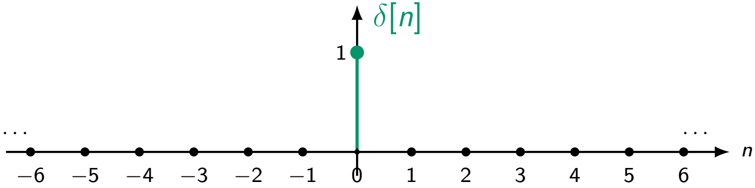
Graph of $u[n]$



The discrete-time unit impulse is nonzero only at the origin

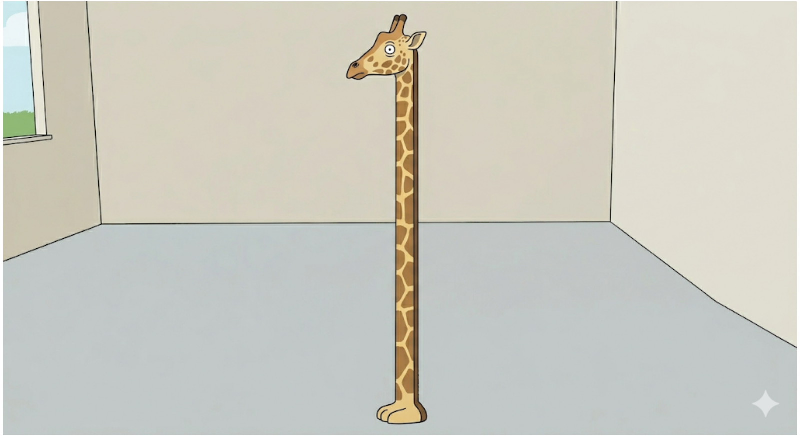
$$\delta[n] = \begin{cases} 1, & n = 0, \\ 0, & n \neq 0. \end{cases}$$

Graph of $\delta[n]$



Unit step function: The sleeping giraffe

Unit impulse function: The giraffe wakes up



Practice

Use the definitions of $u[n]$ and $\delta[n]$.

Unit step

$$
\begin{aligned}
\text{(a)} \quad u[1] &= ? \\
\text{(b)} \quad u[-3] &= ? \\
\text{(c)} \quad u[0] &= ?
\end{aligned}
$$

Unit impulse

$$
\begin{aligned}
\text{(d)} \quad \delta[0] &= ? \\
\text{(e)} \quad \delta[4] &= ? \\
\text{(f)} \quad \delta[-2] &= ?
\end{aligned}
$$

Answers: values of $u[n]$ and $\delta[n]$

Unit step

$$\begin{aligned} u[1] &= 1 \\ u[-3] &= 0 \\ u[0] &= 1 \end{aligned}$$

Since $u[n] = 1$ for $n \geq 0$.

Unit impulse

$$\begin{aligned} \delta[0] &= 1 \\ \delta[4] &= 0 \\ \delta[-2] &= 0 \end{aligned}$$

Since $\delta[n] = 1$ only at $n = 0$.

Practice: how do shifted signals look?

Can we plot each of the signals below?

Shifted steps

(a) $u[n-2] = ?$

(b) $u[n+3] = ?$

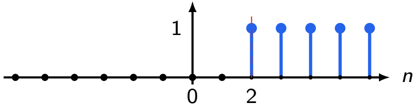
Shifted impulses

(c) $\delta[n-2] = ?$

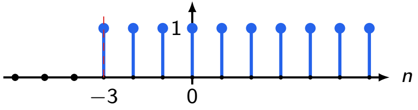
(d) $\delta[n+3] = ?$

Answers: shifted steps and shifted impulses

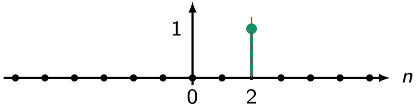
$u[n-2]$



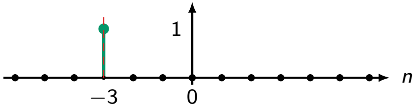
$u[n+3]$



$\delta[n-2]$

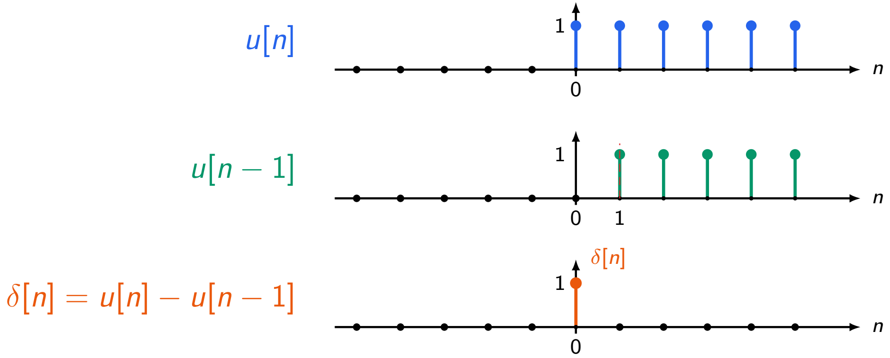


$\delta[n+3]$



R1: The impulse is the first difference of the step function

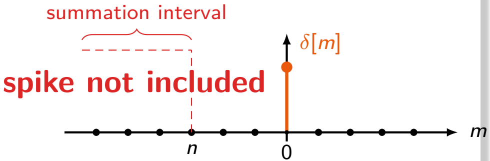
$$\delta[n] \quad = \quad u[n] \;-\; u[n \;-\; 1]$$



R2: A running sum asks whether the impulse is included

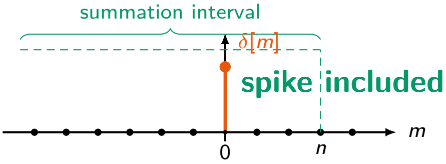
$$\sum_{m=-\infty}^{n} \delta[m]$$

Case 1: $n < 0$



$$\sum_{m=-\infty}^{n} \delta[m] = 0$$

Case 2: $n \geq 0$



$$\sum_{m=-\infty}^{n} \delta[m] = 1$$

R2: The running sum of the impulse is the step function

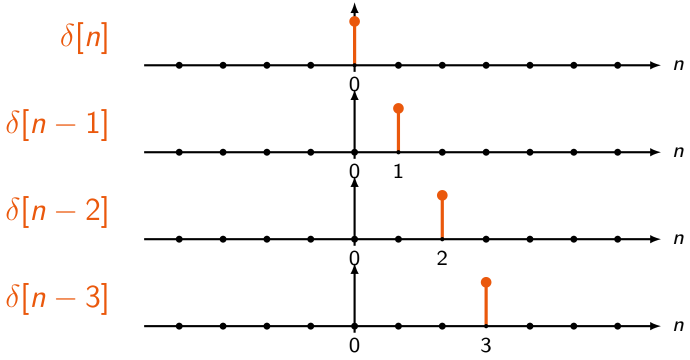
$$u[n] \quad = \quad \sum_{m=-\infty}^{n} \delta[m]$$

Why?

$$\begin{aligned} n < 0 \quad &\Rightarrow \quad \text{the spike at } m = 0 \text{ is not included} \\ &\Rightarrow \quad 0 \\ n \geq 0 \quad &\Rightarrow \quad \text{the spike at } m = 0 \text{ is included} \\ &\Rightarrow \quad 1 \end{aligned}$$

R3: Delayed impulses can build the step function

Each delayed impulse $\delta[n-k]$ contributes exactly one sample, at $n = k$.



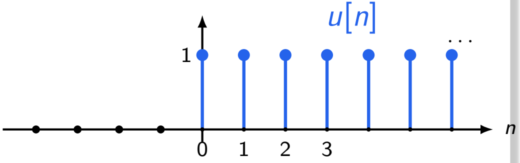
R3: Summing delayed impulses gives the unit step function

$$u[n] = \delta[n] + \delta[n-1] + \delta[n-2] + \cdots = \sum_{k=0}^{\infty} \delta[n-k]$$

Interpretation

$$\begin{aligned} \delta[n] &\Rightarrow \text{sample at } n = 0 \\ \delta[n-1] &\Rightarrow \text{sample at } n = 1 \\ \delta[n-2] &\Rightarrow \text{sample at } n = 2 \\ \vdots &\Rightarrow \vdots \end{aligned}$$

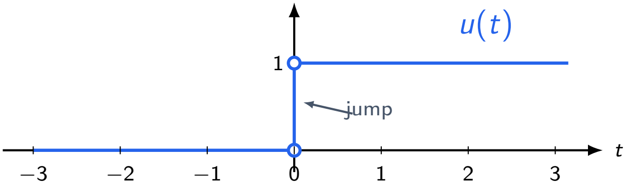
The sum: unit step function



The continuous-time unit step jumps at $t = 0$

$$u(t) = \begin{cases} 0, & t < 0, \\ 1, & t > 0. \end{cases}$$

Graph of $u(t)$



The value at exactly $t = 0$ is usually not important for our signal operations.

The impulse should be the derivative of the step

Discrete time gave us

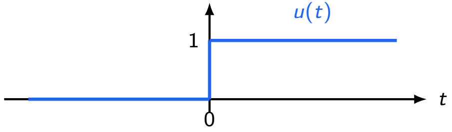
$$\delta[n] = u[n] - u[n-1].$$

This looks like a difference. In continuous time, we want the analogous idea:

$$\delta(t) = \frac{du(t)}{dt}.$$

The impulse should be the derivative of the step

Step



Derivative idea

$$\frac{d}{dt}u(t)$$

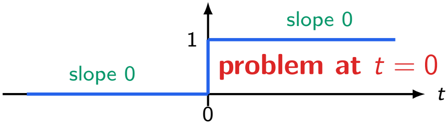
The function is flat every-where except at the jump.

The jump should be-come an impulse.

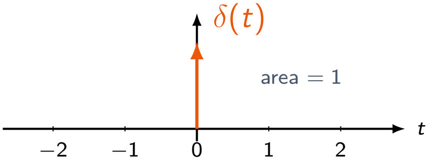
But Oops, we can't differentiate at the origin!

Away from $t = 0$, the step is constant, so derivative is 0. At $t = 0$, the step jumps instantly, so the ordinary derivative is not defined.

Ordinary derivative



Impulse notation



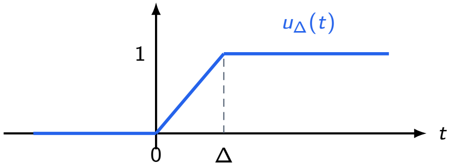
An arrow represents an ideal impulse.

(*) The derivative of a smoothed step is a tall narrow pulse

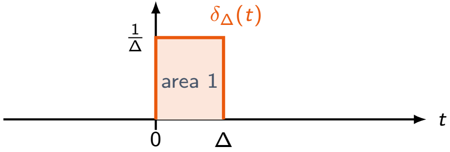
We first smooth the jump over a tiny interval Δ, then take the derivative.

$$\delta_\Delta(t) = \frac{du_\Delta(t)}{dt}$$

Approximate step $u_\Delta(t)$



Derivative $\delta_\Delta(t)$



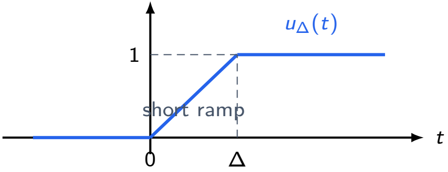
(*) We first replace the instant jump by a very short ramp

Instead of jumping instantly, let the signal rise from 0 to 1 over a short interval Δ.

$$u_\Delta(t) = \begin{cases} 0, & t < 0, \\ \dfrac{t}{\Delta}, & 0 \le t \le \Delta, \\ 1, & t > \Delta. \end{cases}$$

(*) We first replace the instant jump by a very short ramp

Smoothed step $u_\Delta(t)$

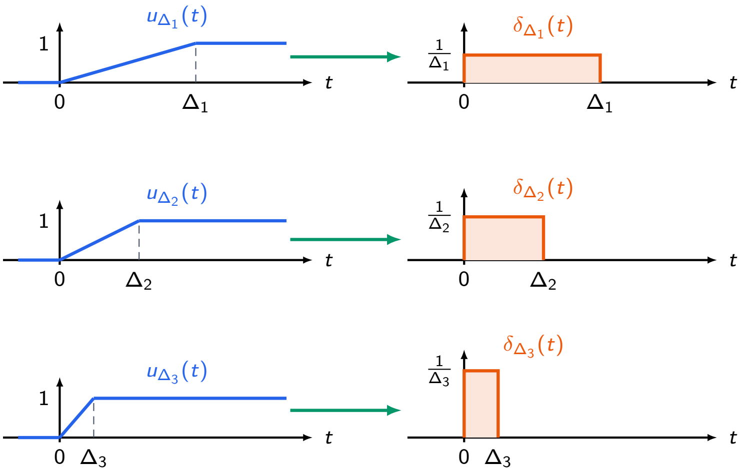


Its derivative

$$\delta_\Delta(t) = \frac{du_\Delta(t)}{dt} = \begin{cases} \frac{1}{\Delta}, & 0 < t < \Delta, \\ 0, & \text{otherwise.} \end{cases}$$

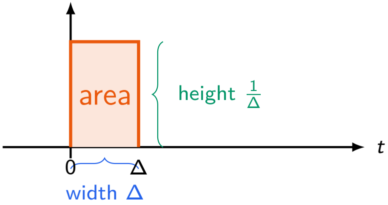
The steeper the ramp, the taller the derivative pulse.

(*) Making the transition shorter makes the pulse taller



(*) The area under the pulse stays equal to 1

Area of $\delta_\Delta(t)$



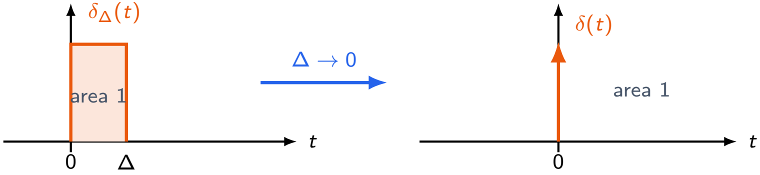
So the area is

$$\int_{-\infty}^{\infty} \delta_\Delta(t)\, dt = \Delta \cdot \frac{1}{\Delta}$$
$$= 1.$$

This is the quantity that survives as the pulse becomes ideal.

The ideal impulse is the limiting infinitely narrow pulse

From $\delta_\Delta(t)$ to $\delta(t)$

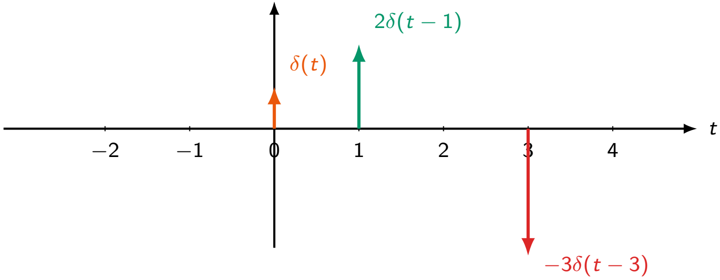


$$\delta(t) = \frac{du(t)}{dt}, \qquad u(t) = \int_{-\infty}^{t} \delta(\tau)\, d\tau.$$

How to draw impulses with arrows

The arrow height is not the actual value of the impulse. The number beside the arrow just tells us the **area** or **strength**.

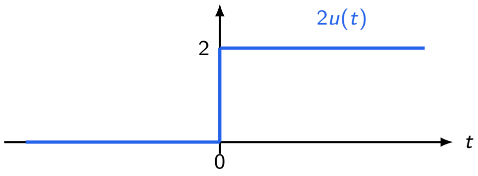
Examples of impulse arrows



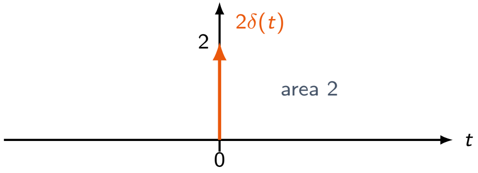
Example: differentiating $2u(t)$

$$\frac{d}{dt}\left[2u(t)\right] = 2\delta(t)$$

Signal: $2u(t)$



Derivative: $2\delta(t)$



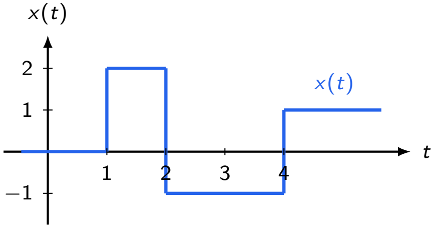
Practice: write the signal using shifted steps

Observe the jumps

$$
\begin{aligned}
t = 1 &: \quad 0 \to 2 \quad \Rightarrow \quad +2 \\
t = 2 &: \quad 2 \to -1 \quad \Rightarrow \quad -3 \\
t = 4 &: \quad -1 \to 1 \quad \Rightarrow \quad +2
\end{aligned}
$$

$$x(t) = 2u(t-1) - 3u(t-2) + 2u(t-4)$$

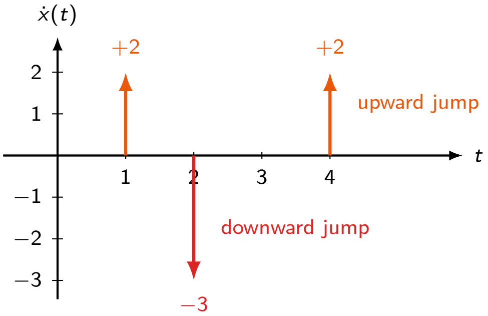
Signal $x(t)$



Practice: differentiate the jump signal (Example 1.7)

$$\frac{dx(t)}{dt} = 2\delta(t-1) - 3\delta(t-2) + 2\delta(t-4)$$

Derivative $\dot{x}(t)$



Exponential and Sinusoidal Signals

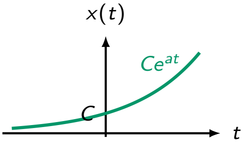
A real exponential either grows, decays, or stays flat

The basic continuous-time real exponential has the form

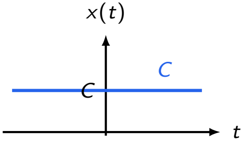
$$x(t) = Ce^{at},$$

where C and a are real constants.

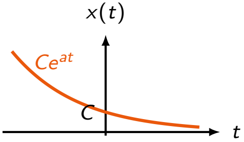
$a > 0$: **grows**



$a = 0$: **constant**



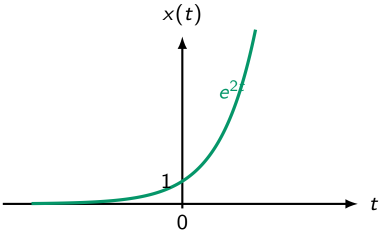
$a < 0$: **decays**



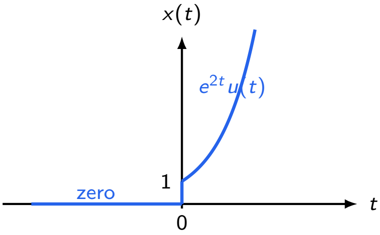
Multiplying by a step function keeps only part of a signal

The unit step acts like **a switch**:
$x(t)u(t)$ keeps the right side of $x(t)$ and removes the left side.

Original exponential e^{2t}

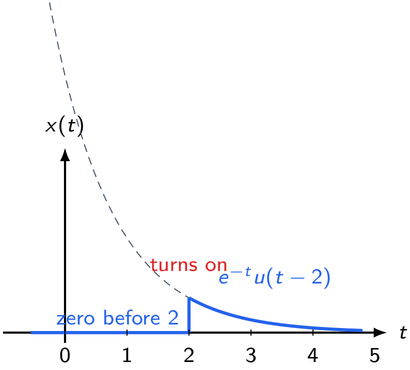


After multiplying by $u(t)$



Shifted steps choose where the exponential begins

Graph of $e^{-t}u(t-2)$



Practice: what do these exponential-step signals look like?

Left/right selection

$$\text{(a)} \quad e^{t}u(-t) = ?$$

$$\text{(b)} \quad e^{-t}u(t-2) = ?$$

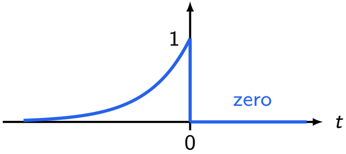
Shifted switches

$$\text{(c)} \quad e^{t}u(3-t) = ?$$

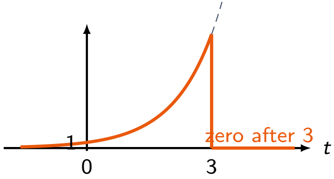
$$\text{(d)} \quad e^{-t}u(3t) = ?$$

Remember: $u(3t) = u(t)$, because $3t$ has the same sign as t.

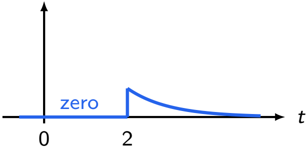
$$e^{t}u(-t)$$



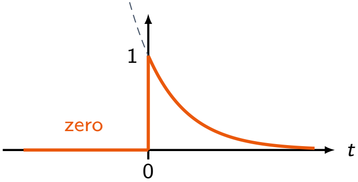
$$e^{t}u(3-t)$$



$$e^{-t}u(t-2)$$



$$e^{-t}u(3t) = e^{-t}u(t)$$

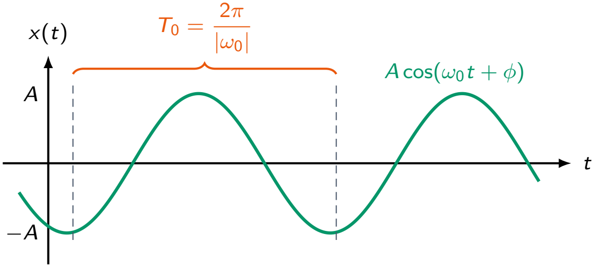


A sinusoid repeats after one full cycle

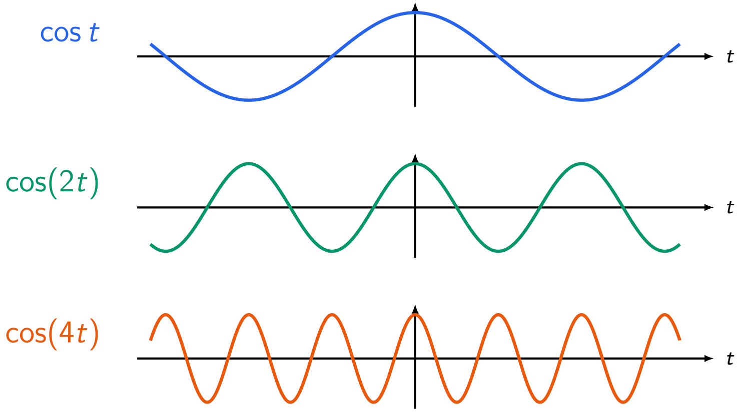
$$x(t) \quad = \quad A\cos(\omega_0 t \; + \; \phi).$$

A controls height, ω_0 controls speed of oscillation, and ϕ shifts the wave.

Period of a continuous-time sinusoid

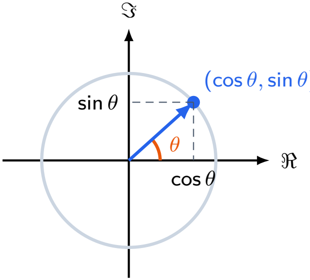


Larger $|\omega_0|$ means faster oscillation



A rotating point carries both cosine and sine

Unit circle view



Angle $\theta = \omega_0 t$ gives two signals

$\cos(\omega_0 t)$ and $\sin(\omega_0 t)$

They are the horizontal and vertical coordinates of the same rotating point.

Euler's formula packages two sinusoids into one expression

$$e^{j\theta} = \cos\theta + j\sin\theta$$

With $\theta = \omega_0 t$

$$e^{j\omega_0 t} = \cos(\omega_0 t) + j\sin(\omega_0 t).$$

One compact expression carries both cosine and sine.

Real and imaginary parts

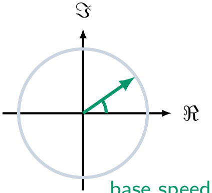
$$\Re\{e^{j\omega_0 t}\} = \cos(\omega_0 t)$$

$$\Im\{e^{j\omega_0 t}\} = \sin(\omega_0 t)$$

Changing ω_0 changes how fast the arrow rotates

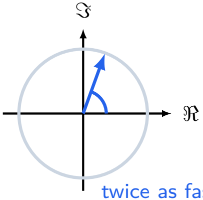
The signal $e^{j\omega_0 t}$ moves around the unit circle.
Larger $|\omega_0|$ means faster rotation.

$e^{j\omega_0 t}$



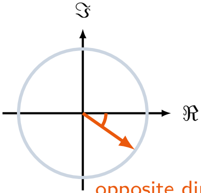
base speed

$e^{j2\omega_0 t}$



twice as fast

$e^{-j\omega_0 t}$

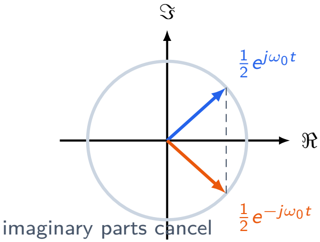


opposite direction

A real cosine is the sum of two opposite rotations

$$\cos(\omega_0 t) = \frac{1}{2}e^{j\omega_0 t} + \frac{1}{2}e^{-j\omega_0 t}$$

Two rotating arrows



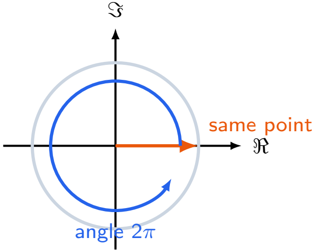
What remains is real

$$e^{j\theta} + e^{-j\theta} = 2\cos\theta,$$

$$e^{j\theta} - e^{-j\theta} = 2j\sin\theta.$$

Continuous complex exponentials repeat after a full rotation

One full rotation



Period condition

$$e^{j\omega_0(t+T)} = e^{j\omega_0 t}$$

$$e^{j\omega_0 t} e^{j\omega_0 T} = e^{j\omega_0 t}$$

$$e^{j\omega_0 T} = 1$$

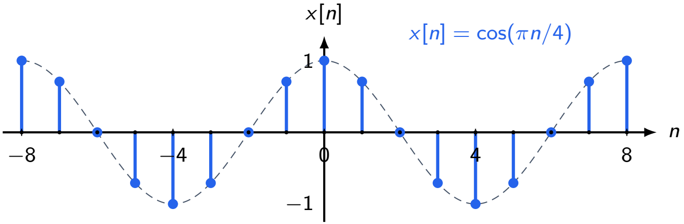
$$\omega_0 T = 2\pi \qquad \Rightarrow \qquad T_0 = \frac{2\pi}{|\omega_0|}.$$

A discrete periodic signal: A sampled cosine

In discrete time, we only keep the values at integer indices:

$$x[n] = \cos(\omega_0 n).$$

Samples of $\cos\left(\frac{\pi}{4}t\right)$



Discrete-time periodicity asks for an integer shift

Discrete time periodicity

$$\cos(\omega_0 n)$$

$$\text{repeats after every multiple of } \frac{2\pi}{\omega_0}$$

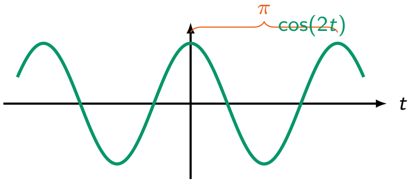
$$\text{But, some multiple } N = m\frac{2\pi}{\omega_0} \quad \text{must be an integer.}$$

The discrete signal is periodic **only if** such an integer N exists.

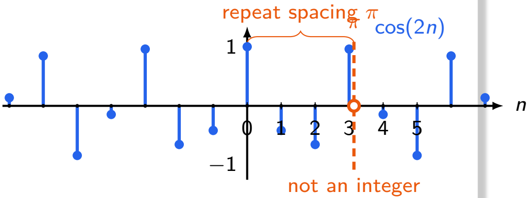
Example: $\cos(2t)$ is periodic, but $\cos(2n)$ is not

For the continuous signal $x(t) = \cos(2t)$, the period, $T_0 = \frac{2\pi}{2} = \pi$.
But π is not an integer shift in discrete time.

Continuous-time



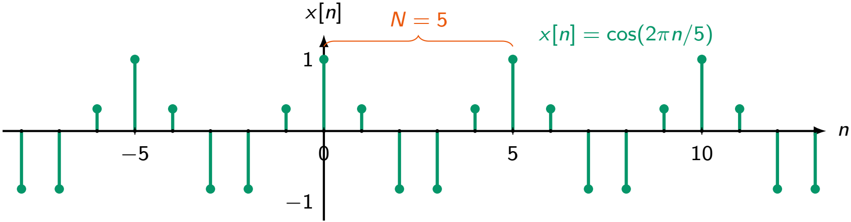
Discrete-time



Example: $\cos\left(\frac{2\pi}{5}n\right)$ repeats every 5 samples

After $N = 5$ samples, the angle has increased by 2π, so it repeats.

A periodic discrete-time sinusoid



Practice: is $3\sin\left(\frac{5}{7}n\right)$ periodic?

$$x[n] \quad = \quad 3\sin\left(\tfrac{5}{7}n\right)$$

Question

Is there a positive integer N so that

$$x[n+N] = x[n]$$

for every integer n?

Hint

We need

$$\frac{2\pi}{\omega_0} = \frac{N}{m}$$

Answer: $3\sin\left(\frac{5}{7}n\right)$ is not periodic

$$x[n] \quad = \quad 3\sin\left(\tfrac{5}{7}n\right) \qquad \text{is not periodic.}$$

Apply the test

$$\frac{2\pi}{\omega_0} = \frac{2\pi}{5/7} = \frac{14\pi}{5} = \frac{N}{m}$$

But $\frac{N}{m}$ must be rational

Why it fails

For $m \quad \neq \quad 0$,

$$\frac{14\pi}{5}m$$

is irrational, so it can never be an integer N.

Practice: find the period if it exists

$$x[n] \quad = \quad 2\cos\left(\frac{16\pi}{63}n + \frac{\pi}{3}\right)$$

Hint

The phase shift

$$\frac{\pi}{3}$$

does not affect the period.

Only the coefficient of n matters.

Answer: the fundamental period is $N_0 = 63$

$$x[n] \quad = \quad 2\cos\left(\tfrac{16\pi}{63}n + \tfrac{\pi}{3}\right)$$

Find when the samples line up again

$$\frac{2\pi}{\omega_0} = \frac{2\pi}{16\pi/63} = \frac{63}{8} = \frac{N}{m}$$

So after 8 cycles, the samples land exactly 63 steps later: $8 \cdot \frac{63}{8} = 63$.

Thank you!